

Time Series

Introduction to Time Series

The statistical data is recorded with its time of occurrence is called a time series. The yearly output of wheat recorded for the last twenty five years, the weekly average price of eggs recorded for the last 52 weeks, the monthly average sales of a firm recorded for the last 48 months or the quarterly average profits recorded for the last 40 quarter etc., are example of time series data. It may be observed that this data undergoes changes with the passage of time. A number of factors can be isolated with contribute to the changes occurring overtime in such series.

In the field of economics and business, for example, income, imports exports, production, consumption, prices these data are depend on time. And all of these data are pretentious by seasonal changes as well as regular cyclical changes over the time period. To evaluate the changes in business and economics, the analysis of time series plays an important role in this regard. It is necessary to associated time with time series because time is one basic variable in time series analysis.

Definition of time series:

A time- series may be defined as a collection of readings belonging to different time periods, of some economic variable or composite of variables.

A time series consists of a set of observations arranged in chronological order (either ascending or descending). Time Series has an important objective to identify the variations and try to eliminate the variations and also helps us to estimate or predict the future values.

Role of time series Analysis

Time series Analysis is of great significance in business decision- making for the following reason:

- (1) ***It helps in the understanding of past behavior:*** By observing data over a period of time, one can easily understand what changes have taken place in the past. Such analysis will be extremely helpful in predicting the future behavior.
- (2) ***It helps in planning future operations:*** Statistical techniques have been evolved which enable time series to be analysed in such a way that the influences which have determined the form of that series may be ascertained. If the regularity of occurrence of any feature over a sufficient long period could be clearly established then, within limits, prediction of probable future variations would become possible.
- (3) ***It helps in evaluating current accomplishments:*** The actual performance can be compared with the expected performance and the cause of variation analyzed. If expected sales for 2011 were 95 lakhs color T.V sets and the actual sale were only 94 lakhs; one can investigate the cause for the shortfall in achievement.

(4) It facilitates comparison: Different time series are often compared and important conclusions drawn there from.

So Time Series means

It helps in the analysis of the past behavior.

It helps in forecasting and for future plans.

It helps in the evaluation of current achievements.

It helps in making comparative studies between one time period and others.

Therefore time series helps us to study and analyze the time related data which involves in business fields, economics, industries, etc...

Series Uses of Time

Time series is used to predict future values based on previously observed values.

Time series analysis is used to identify the fluctuation in economics and business.

It helps in the evaluation of current achievements.

Time series is used in pattern recognition, signal processing, weather forecasting and earthquake prediction.

It can be said that time series analysis is a big tool in the hands of business executives to plan their sales, prices, policies and production.

Components of Time Series

There are four types of components in a time series. They are as follows;

(i) Secular Trend

(ii) Seasonal variations

(iii) Cyclic variations

(iv) Irregular variations

(i) Secular Trend

It is a general tendency of time series to increase or decrease or stagnates during a long period of time. An upward tendency is usually observed in population of a country, production, sales, prices in industries, income of individuals etc., A downward tendency is observed in deaths, epidemics, prices of electronic gadgets, water sources, mortality rate etc.... It is not necessarily that the increase or decrease should be in the same direction throughout the given period of time.

(ii) Seasonal Variations

As the name suggests, tendency movements are due to nature which repeat themselves periodically in every seasons. These variations repeat themselves in less than one year time. It is measured in an interval of time. Seasonal variations may be influenced by natural force, social customs and traditions. These variations are the results of such factors which uniformly and regularly rise and fall in the magnitude. For example, selling of umbrellas' and raincoat in the rainy season, sales of cool drinks in summer season, crackers in Deepawali season, purchase of dresses in a festival season, sugarcane in Pongal season.

(iii) Cyclic Variations

These variations are not necessarily uniformly periodic in nature. That is, they may or may not follow exactly similar patterns after equal intervals of time. Generally one cyclic period ranges from 7 to 9 years and there is no hard and fast rule in the fixation of years for a cyclic period. For example, every business cycle has a Start-Boom- Depression-Recover, maintenance during booms and depressions, changes in government monetary policies, changes in interest rates.

(iv) Irregular Variations

These variations do not have particular pattern and there is no regular period of time of their occurrences. These are accidently changes which are purely random or unpredictable. Normally they are short-term variations, but its occurrence sometimes has its effect so intense that they may give rise to new cyclic or other

movements of variations. For example floods, wars, earthquakes, Tsunami, strikes, lockouts etc...

Mathematical Model for a Time Series

There are two common models used for decomposition of a time series into its components, namely additive and multiplicative model.

(i) Additive Model:

This model assumes that the observed value is the sum of all the four components of time series. (i.e) $Y = T + S + C + I$

where Y = Original value , T = Trend Value , S = Seasonal component

C = Cyclic component , I = Irregular component

The additive model assumes that all the four components operate independently. It also assumes that the behavior of components is of an additive character.

(ii) Multiplicative Model:

This model assumes that the observed value is obtained by multiplying the trend(T) by the rates of other three components. $Y = T \times S \times C \times I$

where Y = Original value , T = Trend Value , S = Seasonal component

C = Cyclic component , I = Irregular component

This model assumes that the components due to different causes are not necessarily independent and they can affect one another. It also assumes that the behavior of components is of a multiplicative character.

Measurements of Trends

Following are the methods by which we can measure the trend.

(i) Freehand or Graphic Method.

- (ii) Method of Semi-Averages.
- (iii) Method of Moving Averages.
- (iv) Method of Least Squares.

(i) Freehand or Graphic Method.

It is the simplest and most flexible method for estimating a trend. We will see the working procedure of this method.

Procedure:

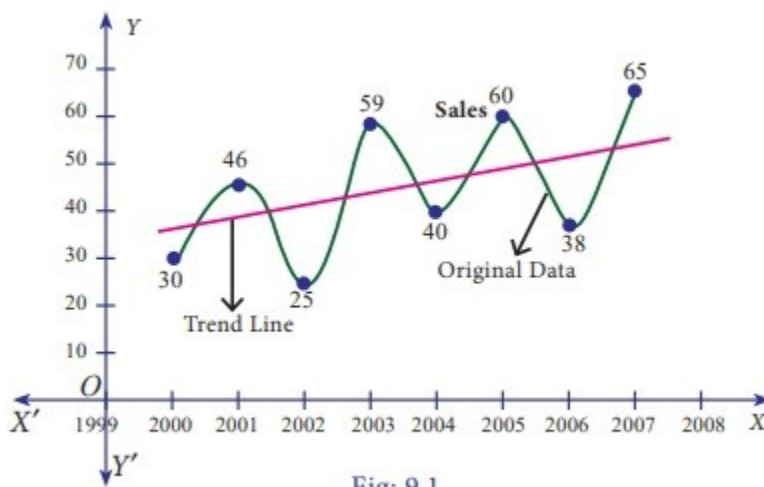
- (a) Plot the time series data on a graph.
- (b) Draw a freehand smooth curve joining the plotted points.
- (c) Examine the direction of the trend based on the plotted points.
- (d) Draw a straight line which will pass through the maximum number of plotted points.

Example 9.1

Fit a trend line by the method of freehand method for the given data.

Year	2000	2001	2002	2003	2004	2005	2006	2007
Sales	30	46	25	59	40	60	38	65

Solution:



Note

The trend drawn by the freehand method can be extended to predict the future values of the given data. However, this method is subjective in nature, predictions

obtained by this method depends on the personal bias and judgement of the investigator handling the data.

(ii) Method of Semi-Averages

In this method, the semi-averages are calculated to find out the trend values. Now, we will see the working procedure of this method.

Procedure:

- (i) The data is divided into two equal parts. In case of odd number of data, two equal parts can be made simply by omitting the middle year.
- (ii) The average of each part is calculated, thus we get two points.
- (iii) Each point is plotted at the mid-point (year) of each half.
- (iv) Join the two points by a straight line.
- (v) The straight line can be extended on either side.
- (vi) This line is the trend line by the methods of semi-averages.

Example 9.2

Fit a trend line by the method of semi-averages for the given data.

Year	2000	2001	2002	2003	2004	2005	2006
Production	105	115	120	100	110	125	135

Solution:

Since the number of years is odd(seven), we will leave the middle year's production value and obtain the averages of first three years and last three years.

Year	Production	Average
2000	105	$\frac{105 + 115 + 120}{3} = 113.33$
2001	115	
2002	120	
2003	100 (left out)	
2004	110	$\frac{110 + 125 + 135}{3} = 123.33$
2005	125	
2006	135	

Table 9.1

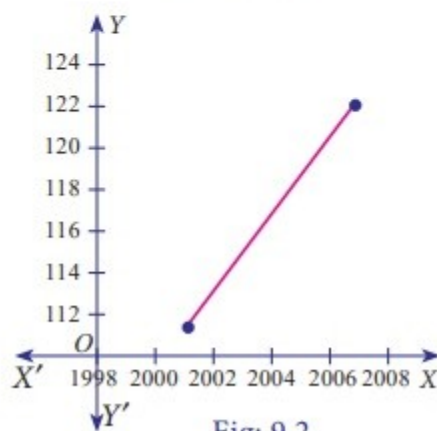


Fig: 9.2

Example 9.3

Fit a trend line by the method of semi-averages for the given data.

Year	1990	1991	1992	1993	1994	1995	1996	1997
Sales	15	11	20	10	15	25	35	30

Solution:

Since the number of years is even(eight), we can equally divide the given data into two equal parts and obtain the averages of first four years and last four years.

Year	Production	Average
1990	15	$\frac{15 + 11 + 20 + 10}{4} = 14$
1991	11	
1992	20	
1993	10	
1994	15	$\frac{15 + 25 + 35 + 30}{4} = 26.25$
1995	25	
1996	35	
1997	30	

Table 9.2

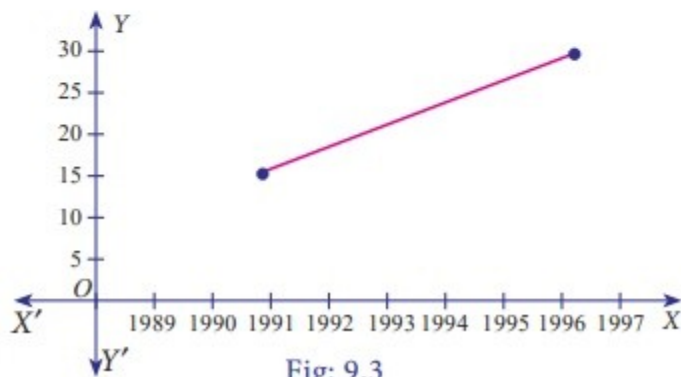


Fig: 9.3

Note

- (i) The future values can be predicted.
- (ii) The trend values obtained by this method and the predicted values are not precise.

(iii) Method of Moving Averages

Moving Averages Method gives a trend with a fair degree of accuracy. In this method, we take arithmetic mean of the values for a certain time span. The time span can be three-years, four -years, five- years and so on depending on the data set and our interest. We will see the working procedure of this method.

Procedure:

(i) Decide the period of moving averages (three- years, four -years).

(ii) In case of odd years, averages can be obtained by calculating,

$$\frac{a+b+c}{3}, \frac{b+c+d}{3}, \frac{c+d+e}{3}, \frac{d+e+f}{3}, \dots$$

(iii) If the moving average is an odd number, there is no problem of centering it, the average value will be centered besides the second year for every three years.

(iv) In case of even years, averages can be obtained by calculating,

$$\frac{a+b+c+d}{4}, \frac{b+c+d+e}{4}, \frac{c+d+e+f}{4}, \frac{d+e+f+g}{4}, \dots$$

(v) If the moving average is an even number, the average of first four values will be placed between 2nd and 3rd year, similarly the average of the second four values will be placed between 3rd and 4th year. These two averages will be again averaged and placed in the 3rd year. This continues for rest of the values in the problem. This process is called as centering of the averages.

Example 9.4

Calculate three-yearly moving averages of number of students studying in a higher secondary school in a particular village from the following data.

Year	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004
Number of students	332	317	357	392	402	405	410	427	435	438

Solution:

Computation of three- yearly moving averages.

Year	Number of students	3- yearly moving Total	3- yearly moving Averages
1995	332	---	---
1996	317	1006	335.33
1997	357	1066	355.33
1998	392	1151	383.67
1999	402	1199	399.67
2000	405	1217	405.67
2001	410	1242	414.00
2002	427	1272	424.00
2003	435	1300	433.33
2004	438	---	---

Table 9.3

Example 9.5

Calculate four-yearly moving averages of number of students studying in a higher secondary school in a particular city from the following data.

Year	2001	2002	2003	2004	2005	2006	2007	2008
Sales	124	120	135	140	145	158	162	170

Solution:

Computation of four- yearly moving averages.

Year	Sales	4-yearly centered moving total	4-yearly moving Average	4-yearly centered moving Average
2001	124	--	--	--
2002	120	--	--	--
		519	129.75	
2003	135	--		132.37
		540	135.00	
2004	140	--		139.75
		578	144.50	
2005	145	--		147.87
		605	151.25	
2006	158	--		155.00
		635	158.75	
2008	162	--		162.50
		665	166.25	
2007	170	--	--	-
2008	175	--	--	-

Table 9.4

Note

The calculated 4-yearly centered moving average belongs to the particular year present in that row eg; 132.37 belong to the year 2003.

(iv) Method of Least Squares

The line of best fit is a line from which the sum of the deviations of various points is zero. This is the best method for obtaining the trend values. It gives a convenient basis for calculating the line of best fit for the time series. It is a mathematical method for measuring trend. Further the sum of the squares of these deviations would be least when compared with other fitting methods. So, this method is known as the Method of Least Squares and satisfies the following conditions:

(i) The sum of the deviations of the actual values of Y and $\hat{Y}$ (estimated value of Y) is Zero. that is $\Sigma(Y - \hat{Y}) = 0$.

(ii) The sum of squares of the deviations of the actual values of Y and $\hat{Y}$ (estimated value of Y) is least. that is $\Sigma(Y - \hat{Y})^2$ is least ;

Procedure:

(i) The straight line trend is represented by the equation $Y = a + b X$...[\(1\)](#)

where Y is the actual value, X is time, a , b are constants

(ii) The constants 'a' and 'b' are estimated by solving the following two normal

Equations $\Sigma Y = n a + b \Sigma X$...[\(2\)](#)

$$\Sigma XY = a \Sigma X + b \Sigma X^2 \quad \dots\text{[\(3\)](#)}$$

Where 'n' = number of years given in the data.

(iii) By taking the mid-point of the time as the origin, we get $\Sigma X = 0$

(iv) When $\Sigma X = 0$, the two normal equations reduces to

$$\Sigma Y = n a + b (0) \quad ; \quad a = \frac{\Sigma Y}{n} = \bar{Y}$$
$$\Sigma XY = a(0) + b \Sigma X^2 \quad ; \quad b = \frac{\Sigma XY}{\Sigma X^2}$$

The constant 'a' gives the mean of Y and 'b' gives the rate of change (slope).

(v) By substituting the values of 'a' and 'b' in the trend equation [\(1\)](#), we get the Line of Best Fit.

Example 9.6

Given below are the data relating to the production of sugarcane in a district.

Fit a straight line trend by the method of least squares and tabulate the trend values.

Year	2000	2001	2002	2003	2004	2005	2006
Prod. of Sugarcane	40	45	46	42	47	50	46

Solution:

Computation of trend values by the method of least squares (ODD Years).

Year(x)	Production of Sugarcane(Y)	$X=(x-2003)$	X^2	XY	Trend values(Y_t)
2000	40	-3	9	-120	42.04
2001	45	-2	4	-90	43.07
2002	46	-1	1	-46	44.11
2003	42	0	0	0	45.14
2004	47	1	1	47	46.18
2005	50	2	4	100	47.22
2006	46	3	9	138	48.25
$N= 7$	$\Sigma Y = 316$	$\Sigma X = 0$	$\Sigma X^2= 28$	$\Sigma XY=29$	$\Sigma Y_t = 316$

Table 9.5

$$a = \frac{\sum Y}{n} = \frac{316}{7} = 45.143 ; b = \frac{\sum XY}{\sum X^2} = \frac{29}{28} = 1.036$$

Therefore, the required equation of the straight line trend is given by

$$Y = a + b X;$$

$$Y = 45.143 + 1.036X \quad (x-2003)$$

The trend values can be obtained by

$$\text{When } X = 2000, Y_t = 45.143 + 1.036(2000-2003) = 42.035$$

$$\text{When } X = 2001, Y_t = 45.143 + 1.036(2001-2003) = 43.071,$$

similarly other values can be obtained.

Problem 15

The following are annual profits (in thousand Taka) in a tea stall:

Year	2006	2007	2008	2009	2010	2011	2012
Profit	80	90	92	83	94	99	92

- Use the method of least square to fit a straight line to the above data.
- Estimate likely profit in 2015.
- Plot the above figures and draw the line.

Solution:

(i) Fitting Straight Line Trend by the Method of Least Squares

Year	Production Y	X	XY	X ²	Trend Values Y _c = 88.43 + 5.04X
2006	80	- 3	- 240	9	73.31
2007	90	- 2	- 180	4	78.35
2008	92	- 1	- 92	1	83.39
2009	83	0	0	0	88.43
2010	94	+ 1	+ 94	1	93.47
2011	99	+ 2	+ 198	4	98.51
2012	92	+ 3	+ 276	9	103.54
N = 7	ΣY = 630	ΣX = 0	ΣXY = 56	ΣX ² = 28	ΣY _c = 619

$$\text{Where, } a = \frac{\sum Y}{N} = \frac{630}{7} = 90 \quad ; \quad b = \frac{\sum XY}{\sum X^2} = \frac{56}{28} = 2$$

The equation of straight line trend is $Y_c = a + bX$

$$\Rightarrow Y_c = 90 + 2X \text{ (Ans).}$$

(ii) Graph of trend line:

(iii) For the year 2015, $X = 6$.

$$\therefore Y_{2008} = 90 + 2X = 90 + 2(6) = 102 \text{ (Ans)}$$

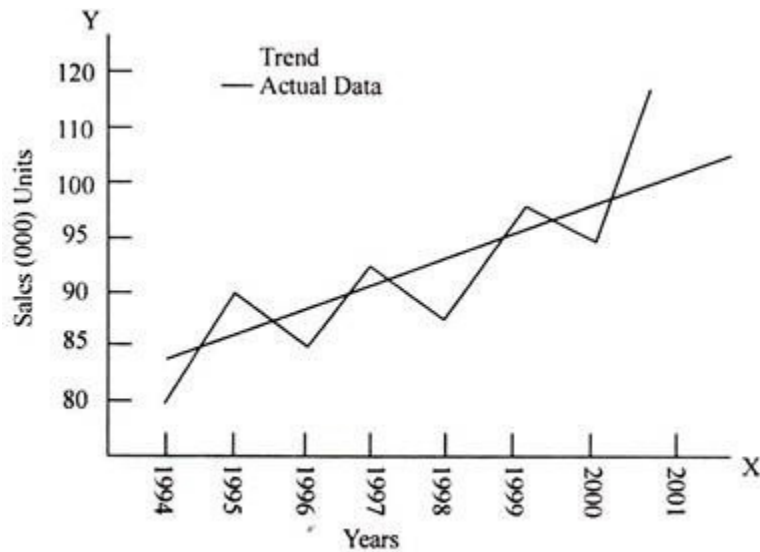


Fig. 2-1. Straight line trend by graphical method.

Example 9.7

Given below are the data relating to the sales of a product in a district.

Fit a straight line trend by the method of least squares and tabulate the trend values.

Year	1995	1996	1997	1998	1999	2000	2001	2002
Sales	6.7	5.3	4.3	6.1	5.6	7.9	5.8	6.1

Solution:

Computation of trend values by the method of least squares.

In case of EVEN number of years, let us consider

$$X = \frac{(x - \text{Arithmetic mean of two middle years})}{0.5}$$

Year(x)	Sales(Y)	$X = \frac{(x - 1998.5)}{0.5}$	XY	X^2	Trend values(Y_t)
1995	6.7	-7	46.9	49	5.6166
1996	5.3	-5	26.5	25	5.7190
1997	4.3	-3	12.9	9	5.8214
1998	6.1	-1	6.1	1	5.9238
1999	5.6	7	39.2	49	6.0261
2000	7.9	5	39.5	25	6.1285
2001	5.8	3	17.4	9	6.2309
2002	6.1	1	6.1	1	6.3333
N=8	47.8	$\Sigma X = 0$	194.6	168	

Table 9.6

$$a = \frac{\sum Y}{n} = \frac{47.8}{8} = 5.975 \quad ; \quad b = \frac{\sum XY}{\sum X^2} = \frac{194.6}{168} = 0.10238$$

Therefore, the required equation of the straight line trend is given by

$$Y = a + bX \quad ; \quad Y = 5.975 + 0.10238 X.$$

$$\text{When } X = 1995, Y_t = 5.975 + 0.10238 \left(\frac{1995 - 1998.5}{0.5} \right) = 5.6166$$

$$\text{When } X = 1996, Y_t = 5.975 + 0.10238 \left(\frac{1996 - 1998.5}{0.5} \right) = 5.7190$$

similarly other values can be obtained.

Note

- (i) Future forecasts made by this method are based only on trend values.
- (ii) The predicted values are more reliable in this method than the other methods.

Seasonal variation

Seasonal variations are fluctuations within a year over different seasons.

Estimation of seasonal variations requires that the time series data are recorded at even intervals such as quarterly, monthly, weekly or daily, depending on the nature of the time series. Changes due to seasons, weather conditions and social customs are the primary causes of seasonal variations. The main objective of the measurement of seasonal variation is to study their effect and isolate them from the trend.

Methods of constructing seasonal indices

There are four methods of constructing seasonal indices.

1. Simple averages method
2. Ratio to trend method
3. Percentage moving average method
4. Link relatives method

Among these, we shall discuss the construction of seasonal index by the first method only.

Method Averages Simple

Under this method, the time series data for each of the 4 seasons (for quarterly data) of a particular year are expressed as percentages to the seasonal average for that year.

The percentages for different seasons are averaged over the years by using simple average.

The resulting percentages for each of the 4 seasons then constitute the required seasonal indices.

Method of calculating seasonal indices

(i) The data is arranged season-wise

(ii) The data for all the 4 seasons are added first for all the years and the seasonal averages for each year is computed.

(iii) The average of seasonal averages is calculated

(i.e., $\text{Grand average} = \frac{\text{Total of seasonal averages}}{\text{number of years}}$).

(iv) The seasonal average for each year is divided by the corresponding grand average and the results are expressed in percentages and these are called seasonal indices.

Example 7.9

Calculate the seasonal indices for the rain fall (in mm) data in Tamil Nadu given below by simple average method

Year	Season			
	I	II	III	IV
2001	118.4	260.0	379.4	70
2002	85.8	185.4	407.1	8.7
2003	129.8	336.5	403.1	12.0
2004	283.4	360.7	472.1	14.3
2005	231.7	308.5	828.8	15.9

Solution:

Year	Season			
	I	II	III	IV
2001	118.4	260.0	379.4	70
2002	85.8	185.4	407.1	8.7
2003	129.8	336.5	403.1	12.0
2004	283.4	360.7	472.1	14.3
2005	231.7	308.5	828.8	15.9
Seasonal total	849.1	1451.1	2490.5	120.9
Seasonal average	169.82	290.22	498.1	24.18
Seasonal index	69	118	203	10

$$\begin{aligned}
 \text{Grand Average} &= \frac{\text{Total of seasonal averages}}{4} \\
 &= \frac{169.82 + 290.22 + 498.1 + 24.18}{4} \\
 &= \frac{982.32}{4} = 245.58
 \end{aligned}$$

$$\text{Seasonal Index} = \frac{\text{Seasonal average}}{\text{Grand average}} \times 100$$

$$\text{Seasonal Index for Season I} = \frac{169.82}{245.58} \times 100 = 69.15 \approx 69$$

$$\text{Seasonal Index for Season II} = \frac{290.22}{245.58} \times 100 = 118.18 \approx 118$$

$$\text{Seasonal Index for Season III} = \frac{498.1}{245.58} \times 100 = 202.83 \approx 203$$

$$\text{Seasonal Index for Season IV} = \frac{24.18}{245.58} \times 100 = 9.85 \approx 10$$

Example 7.10

Obtain the seasonal indices for the rain fall (in mm) data in India given in the following table.

Quarter \ Year	2009	2010	2011	2012
I	38.2	38.5	55	50.5
II	166.8	250.9	277.7	197
III	612.6	773.1	717.8	706.1
IV	72.2	153.1	65.8	101.1

Solution:

Year	Quarter			
	I	II	III	IV
2009	38.2	166.8	612.6	72.2
2010	38.5	250.9	773.1	153.1
2011	55	277.7	717.8	65.8
2012	50.5	197	706.1	101.1
Seasonal total	182.2	892.4	2809.6	392.2
Seasonal average	45.55	223.1	702.4	98.05
Seasonal index	17	83	263	37

$$\begin{aligned}
 \text{Grand Average} &= \frac{\text{Total of seasonal averages}}{4} \\
 &= \frac{45.55 + 223.1 + 702.4 + 98.05}{4} \\
 &= \frac{1069.10}{4} = 267.28
 \end{aligned}$$

$$\text{Seasonal Index} = \frac{\text{Seasonal average}}{\text{Grand average}} \times 100$$

$$\text{Seasonal Index for Quarter I} = \frac{45.55}{267.28} \times 100 = 17.04 \approx 17$$

$$\text{Seasonal Index for Quarter II} = \frac{223.1}{267.28} \times 100 = 83.47 \approx 83$$

$$\text{Seasonal Index for Quarter III} = \frac{702.4}{267.28} \times 100 = 262.80 \approx 263$$

$$\text{Seasonal Index for Quarter IV} = \frac{98.05}{267.28} \times 100 = 36.69 \approx 37$$

Example 7.11

Calculate the seasonal index for the monthly sales of a product using the method of simple averages.

Months	Jan	Feb	Mar	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec
Year												
2001	15	41	25	31	29	47	41	19	35	38	40	30
2002	20	21	27	19	17	25	29	31	35	39	30	44
2003	18	16	20	28	24	25	30	34	30	38	37	39

Solution:

Computation of seasonal Indices by method of simple averages.

months	Jan	Feb	Mar	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec
Years												
2001	15	41	25	31	29	47	41	19	35	38	40	30
2002	20	21	27	19	17	25	29	31	35	39	30	44
2003	18	16	20	28	24	25	30	34	30	38	37	39
Monthly Total	53	78	72	78	70	97	100	84	100	115	107	113
Monthly Averages	17.67	26	24	26	23.33	32.33	33.33	28	33.33	38.33	35.67	37.67

Table 9.7

$$\text{S.I for Jan} = \frac{\text{Monthly Average (for Jan)}}{\text{Grand average}} \times 100$$

$$\text{Grand Average} = \frac{355.582}{12} = 29.63$$

$$\text{S.I for Jan} = \frac{17.666}{29.361} \times 100 = 59.62; \quad \text{S.I for Feb} = \frac{26}{29.361} \times 100 = 87.77;$$

Similarly other seasonal index values can be obtained.

Months	Jan	Feb	Mar	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec
Seasonal Index	59.62	87.77	80.99	87.77	78.74	109.12	112.49	94.49	112.49	129.36	120.36	126.89

Example 7.12

Calculate the seasonal index for the quarterly production of a product using the method of simple averages.

Year	I Quarter	II Quarter	III Quarter	IV Quarter
2005	255	351	425	400
2006	269	310	396	410
2007	291	332	358	395
2008	198	289	310	357
2009	200	290	331	359
2010	250	300	350	400

Solution:

Computation of Seasonal Index by the method of simple averages.

Year	I Quarter	II Quarter	III Quarter	IV Quarter
2005	255	351	425	400
2006	269	310	396	410
2007	291	332	358	395
2008	198	289	310	357
2009	200	290	331	359
2010	250	300	350	400
Quarterly Total	1463	1872	2170	2321
Quarterly Averages	243.83	312	361.67	386.83

Table 9.8

$$\text{S.I for I Quarter} = \frac{\text{Average of I quarter}}{\text{Grand average}} \times 100$$

$$\text{Grand Average} = \frac{1304.333}{4} = 326.0833$$

$$\text{S.I for I Q} = \frac{243.8333}{326.0833} \times 100 = 74.77 ;$$

$$\text{S.I for II Q} = \frac{312}{326.0833} \times 100 = 95.68 ;$$

$$\text{S.I for III Q} = \frac{361.6667}{326.0833} \times 100 = 110.91 ;$$

$$\text{S.I for IV Q} = \frac{386.833}{326.0833} \times 100 = 118.63$$

Exercise 9.1

13. Seasonal Indices

	I	II	III	IV
Total	18.6	20.8	18.8	20.8
Average	3.72	4.16	3.76	4.16
Seasonal indices	94.1772	105.3165	95.1899	105.3165

The Grand Average = 3.95

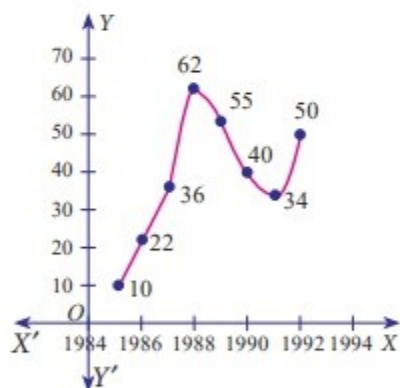
14. Three yearly moving average

Year	1987	1988	1989	1990	1991	1992
Three yearly moving total	46410	52010	63040	79470	94050	102450
Three yearly moving average	15470	17336.666	21013.333	26490	31350	34150

15. Five yearly moving average

Year	1981	1982	1983	1984	1985	1986	1987	1988
Four yearly moving total	619	617	624	621	615	619	613	606
Four yearly moving average	123.8	123.4	124.8	124.2	123	123.8	122.6	121.2

16. Free hand method



17. $a = 169.428$; $b = 3.285$; $Y = 169.428 + 3.285 X$

20. Monthly Indices

	Jan	Feb	Mar	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec
Monthly Total	53	61	54	43	42	51	62	54	52	51	49	67
Monthly Average	17.6	20.3	18	14.3	14	17	20.6	18	17.3	17	16.3	22.3
Seasonal Indices	99.4	114.5	101.4	80.7	78.8	97.1	116.4	101.4	97.6	95.7	92	125.8

Grand Average = 17.74

21.

	I	II	III	IV
Total	372	358	362	364
Average	74.4	71.6	72.4	72.8
Seasonal indices	102.19	98.35	99.45	100

Grand Average = 72.8

22. $a = 46.8$; $b = 3$; $Y = 46.8 + 3 X$

When $X = 1992$, $\hat{Y} = 46.8 + 3 (1992-1994) = 40.8$

When $X = 1993$, $\hat{Y} = 46.8 + 3 (1993-1994) = 43.8$

When $X = 1994$, $\hat{Y} = 46.8 + 3 (1994-1994) = 46.8$

When $X = 1995$, $\hat{Y} = 46.8 + 3 (1995-1994) = 49.8$

When $X = 1996$, $\hat{Y} = 46.8 + 3 (1996-1994) = 52.8$

When $X = 1997$, $\hat{Y} = 46.8 + 3 (1997-1994) = 55.8$

months	Jan	Feb	Mar	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec
Years												
2001	15	41	25	31	29	47	41	19	35	38	40	30
2002	20	21	27	19	17	25	29	31	35	39	30	44
2003	18	16	20	28	24	25	30	34	30	38	37	39
Monthly Total	53	78	72	78	70	97	100	84	100	115	107	113
Monthly Averages	17.67	26	24	26	23.33	32.33	33.33	28	33.33	38.33	35.67	37.67

Table 9.7

$$\text{S.I for Jan} = \frac{\text{Monthly Average (for Jan)}}{\text{Grand average}} \times 100$$

$$\text{Grand Average} = \frac{355.582}{12} = 29.63$$

$$\text{S.I for Jan} = \frac{17.666}{29.361} \times 100 = 59.62; \quad \text{S.I for Feb} = \frac{26}{29.361} \times 100 = 87.77;$$

Similarly other seasonal index values can be obtained.

Months	Jan	Feb	Mar	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec
Seasonal Index	59.62	87.77	80.99	87.77	78.74	109.12	112.49	94.49	112.49	129.36	120.36	126.89

Example 9.9

Calculate the seasonal index for the quarterly production of a product using the method of simple averages.

Year	I Quarter	II Quarter	III Quarter	IV Quarter
2005	255	351	425	400
2006	269	310	396	410
2007	291	332	358	395
2008	198	289	310	357
2009	200	290	331	359
2010	250	300	350	400

Solution:

Computation of Seasonal Index by the method of simple averages.

Year	I Quarter	II Quarter	III Quarter	IV Quarter
2005	255	351	425	400
2006	269	310	396	410
2007	291	332	358	395
2008	198	289	310	357
2009	200	290	331	359
2010	250	300	350	400
Quarterly Total	1463	1872	2170	2321
Quarterly Averages	243.83	312	361.67	386.83

Table 9.8

$$\text{S.I for I Quarter} = \frac{\text{Average of I quarter}}{\text{Grand average}} \times 100$$

$$\text{Grand Average} = \frac{1304.333}{4} = 326.0833$$

$$\text{S.I for I Q} = \frac{243.8333}{326.0833} \times 100 = 74.77 ;$$

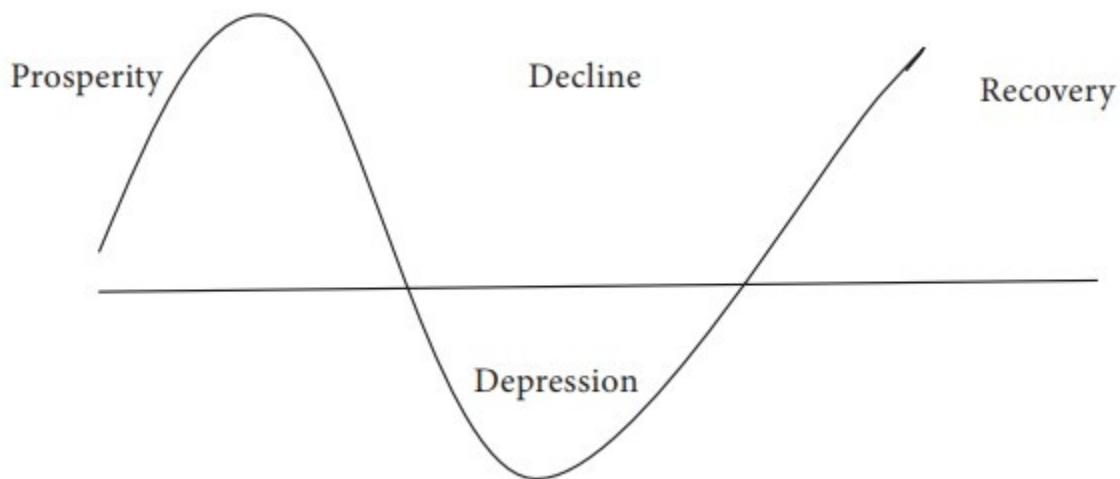
$$\text{S.I for II Q} = \frac{312}{326.0833} \times 100 = 95.68 ;$$

$$\text{S.I for III Q} = \frac{361.6667}{326.0833} \times 100 = 110.91 ;$$

$$\text{S.I for IV Q} = \frac{386.833}{326.0833} \times 100 = 118.63$$

(iii) Cyclical variation

Cyclical variations refer to periodic movements in the time series about the trend line, described by upswings and downswings. They occur in a cyclical fashion over an extended period of time (more than a year). For example, the business cycle may be described as follows.



The cyclical pattern of any time series tells about the prosperity and recession, ups and downs, booms and depression of a business. In most of the businesses there are upward trend for some time followed by a downfall, touching its lowest level. Again a rise starts which touches its peak. This process of prosperity and recession continues and may be considered as a natural phenomenon.

Methods of Cyclical Variation:-

Residual Method

References cycle analysis method

Direct Method

Harmonic Analysis Method

Residual Method:-

Cyclical variations are calculated by Residual Method . This method is based on the multiple model of the time Series.

The process is as below:

(a) When yearly data are given: In class of yearly data there are not any seasonal variations so original data are effect by three components:

- Trend Value
- Cyclical
- Irregular

(b)When monthly or quarterly data are given:

First we calculate the seasonal variation indices according to moving average ratio method.

At last we express the cyclical and irregular variation as the $\rightarrow$ Trend Ratio & Seasonal variation Indices

Measurement of Irregular Variations

The irregular components in a time series represent the residue of fluctuations after trend cycle and seasonal movements have been accounted for. Thus if the original data is divided by T,S and C ; we get I i.e. . In Practice the cycle itself is so erratic and is so interwoven with irregular movement that is impossible to separate them.

